Agentic AI Log Analysis System

Navigation

- Overview
- Upload Logs
- Operational Incident Explorer
- Agent Workflow
- Root Cause Analysis
- Reports & Export
- Settings & History

Quick Start

Generate Demo Logs (5,000)

Load Built-in Sample File

DB Records: 21 sessions | 20 incidents | 10732.0 KB

Multi-Agent Orchestration & ML Trans

Inspect inputs, outputs, execution latencies, and findings for each of the 6 specialized AI agents:

Log Parser Agent (1137.2 ms) — SUCCESS

1. Ingests raw .log, .txt, .csv streams, parses timestamps, levels, services, and endpoints.

Input Summary: Processed file 'sample_logs.log' (5006 total lines, format: Unstructured/Standard)

Output Summary: Successfully parsed 5006 events (0 malformed/skipped lines). Severity breakdown: CRIT 447

Key Findings & Conclusions:

- Parsed 5006 valid operational log entries from sample_logs.log.
- Identified 7 unique applications/services.
- Severity distribution: 185 Critical, 346 Error, 447 Warn, 4028 Info.
- Detected 12 unique API endpoints.

Incident Detection Agent (8838.9 ms) — SUCCESS

2. Heuristic rules + Scikit-Learn TF-IDF classifier to detect anomalies and assign severities.

Input Summary: Scanned 5006 log events across rules and ML models.

Output Summary: Detected 834 anomalous/incident log events. Breakdown by severity: 298 Critical, 496 H

Key Findings & Conclusions:

- Flagged 834 incident candidate events using 7 heuristic rules and TF-IDF ML model.
- Top incident categories: HTTP_SERVER_ERROR (340), SUSPICIOUS_AUTH (135), DATABASE_FAILURE (1
- ML Classifier Performance: Accuracy=70.00%, F1=72.73%, Precision=66.67%.
- Critical-tier events requiring immediate correlation: 298.

Supervised ML Classifier Validation Metrics

Model Accuracy	Precision	Recall
70.00%	66.67%	80.00%

Architecture: TF-IDF + LogisticRegression (Scikit-Learn) | Evaluated on holdout test set with stratified sampling.

Event Correlation Agent (452.0 ms) — SUCCESS

3. Groups related anomalous events using sliding temporal windows, endpoints, and trace links.

Input Summary: Correlated 834 candidate detections against 5006 total events.

Output Summary: Consolidated into 8 distinct operational incidents (8 Critical, 0 High). Found 5 cross-serv

Key Findings & Conclusions:

- Clustered 834 anomaly signals into 8 actionable incidents.
- Average correlated events per incident: 104.2 events.
- Detected 5 cross-service failure cascades (e.g. downstream API failing due to upstream DB).

- Temporal window used for correlation: ±5 minutes.

Root Cause Investigation Agent (9.4 ms) — SUCCESS

4. Reconstructs causal event sequences to formulate evidence-based root cause hypotheses.

Input Summary: Investigated causal telemetry for 8 active incidents.

Output Summary: Formulated 13 evidence-backed root cause hypotheses across 8 incidents (5 High confid uncertainty).

Key Findings & Conclusions:

- Synthesized 13 possible root cause scenarios supported by extracted log evidence lines.
- High confidence hypotheses: 5 (corroborated by clear stack traces or DB connection logs).
- Uncertain/Sparse evidence cases: 8 (telemetry lacks upstream diagnostic context).
- Ethical Observability Standard: All conclusions remain strictly designated as 'Possible Causes' awaiting

Resolution Recommendation Agent (0.1 ms) — SUCCESS

5. Recommends tactical containment steps, diagnostic commands, and preventive runbooks.

Input Summary: Generated troubleshooting runbooks for 8 active incidents.

Output Summary: Prescribed 32 actionable mitigation steps across 8 incidents (16 immediate containment commands).

Key Findings & Conclusions:

- Formulated 32 prioritized troubleshooting recommendations.
- Immediate containment actions: 16 (focused on preventing service cascading failure).
- Included copy-ready shell, SQL, and CLI diagnostic commands for on-call engineers.
- Policy: Actions are recommended guidance; platform requires engineer verification before marking res

Incident Report Agent (2.2 ms) — SUCCESS

6. Compiles comprehensive post-mortem audit reports in TXT, CSV, and JSON formats.

Input Summary: Processed session 'SESS-07E9C72D' containing 8 incidents.

Output Summary: Compiled audit-grade reports across 3 formats (TXT: 22730 chars, CSV: 9 rows, JSON: 52

Key Findings & Conclusions:

- Generated executive text summary with operational metrics and troubleshooting runbooks.
- Produced structured CSV incident registry for ITIL/SIEM record ingestion.
- Generated full JSON artifact containing complete agent findings, evidence, and ML evaluation stats.